Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$NH_4ClO_4 + HNO_3(\text{ dil. }) \longrightarrow HClO_4 + [X]$
$[X] \xrightarrow{\Delta} Y(g)$
$[X]$ and $[Y]$ are respectively $-$

(A) $NH_4NO_3 \& N_2O$

(B) $NH_4NO_2 \& N_2$

(C) $HNO_4 \& O_2$

(D) None of these

Q.2

In nitrogen family, the H-M-Hbond angle in the hydrides gradually becomes closer to 90° on going from N to Sb. This shows that gradually

(A) the basic strength of the hydrides increases.

(B) almost pure p-orbitals are used for $M-H$ bonding.

(C) the bond energies of M -H bonds increases.

(D) the bond pairs of electrons become nearer to the central atom.

Q.3

Brown colour in HNO_3 can be removed by

(A) adding Mg powder.

(B) boiling the acid.

(C) passing NH_3 through acid.

(D) passing air through warm acid.

Q.4

A deep brown gas is formed by mixing two colourless gases which are

(A) NO_2 and O_2

(B) N_2O and NO

(C) NO and O_2

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(D) NH_3 and HCl

Q.5

Which of the following compounds does not exist?

(A) $AsCl_5$

(B) $SbCl_3$

(C) $BiCl_5$

(D) $SbCl_5$

Q.6

The brown ring test for NO_2 and NO_3 is due to the formation of complex ion with a formula

(A) $[Fe(H_2O)_6]^{2+}$

(B) $[Fe(NO)(CN)_5]^{2+}$

(C) $[Fe(H_2O)_5 NO]^{2+}$

(D) $[Fe(H_2O)(NO)_5]^{2+}$

Q.7

An element (X) forms compounds of the formula XCl_3, X_2O_5 and Ca_3X_2 but does not form XCl_5. Which of the following is the element X?

(A) B

(B) Al

(C) N

(D) P

Q.8

Pick out the wrong statement.

(A) Nitrogen has the ability to form $p\pi$ -p π bonds with itself.

(B) Bismuth forms metallic bonds in elemental state.

(C) Catenation tendency is higher in nitrogen when compared with other elements of the same group.

(D) Nitrogen has higher first ionisation enthalpy when compared with other elements of the same group.

Q.9

BCl_3 is a planar molecule whereas NCl_3 is pyramidal because

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(A) BCl_3 has no lone pair of electrons but NCl_3 has a lone pair of electrons.

(B) $B-Cl$ bond is more polar than $N-Cl$ bond.

(C) nitrogen atom is smaller than boron atom.

(D) $N-Cl$ bond is more covalent than $B-Cl$ bond.

Q.10

A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. Y is

(A) $Mg(NO_3)_2$

(B) Mg_3N_2

(C) NH_3

(D) MgO

Q.11

Which one of the following does not have a pyramidal shape?

(A) $(CH_3)_3N$

(B) $(SiH_3)_3N$

(C) $P(CH_3)_3$

(D) $P(SiH_3)_3$

Q.12

Consider the following sequence of reaction.

$$Na + NH_3(g) \longrightarrow [X] \xrightarrow{N_2O} [Y] \xrightarrow{Heat} [Z]$$

Gas Pure

Identify (Z) gas :

(A) N_2

(B) NH_3

(C) O_2

(D) H_2

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Q.13

Among the following compounds, which on heating do not produce N_2 ?

(A) $(NH_4)_2 Cr_2O_7$

(B) $NH_4Cl + NaNO_2$

(C) $NH_4Cl + CaO$

(D) $Ba(N_3)_2$

Q.14

Which of the following statements is not true?

(A) NO_2 can be prepared by heating $Pb(NO_3)_2$.

(B) NO_2 is red - brown gas.

(C) NO_2 is diamagnetic.

(D) NO_2 readily dimerises to N_2O_4.

Q.15

Ammonia, on reaction with hypochlorite anion, can form

(A) NO

(B) NH_4Cl

(C) N_2H_4

(D) HNO_2

Q.16

$(NH_4)_2 Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by

(A) Heating NH_4NO_2

(B) Heating NH_4NO_3

(C) Treating H_2O_2 with $NaNO_2$

(D) Treating $Mg_3 N_2$ with H_2O

Q.17

Among the following elements which one can not form pentahalides.

(A) N

(B) P

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(C) As

(D) Sb

Q.18

When sugar is treated with conc. sulphuric acid, the sugar is charred. In this process, sugar is

(A) reduced

(B) oxidized

(C) sulphonated

(D) dehydrated

Q.19

Which of the following nitrogen oxide is a neutral oxide?

(A) N_2O_3

(B) N_2O_5

(C) NO_2

(D) N_2O

Q.20

(I) $LiNO_3 \xrightarrow{\Delta} X\left(\text{Solid}\right) + \text{gas} + O_2$

(II) $NaNO_3 \xrightarrow[800°C]{\Delta} Y\left(\text{Solid}\right) + \text{gas} + O_2$

The incorrect statement regarding above reactions is

(A) X and Y are respective metal oxides

(B) NO_2 is formed in (II) reaction

(C) Both are redox reactions

(D) NO_2 is formed in (I) reaction

Answers & Solutions

Q.1 Answer:

NH_4NO_3 & N_2O

Solution:

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$$NH_4ClO_4 + HNO_3 \rightarrow \underset{(X)}{HClO_4 + NH_4NO_3}$$

$$NH_4NO_3 \xrightarrow{\Delta} \underset{(Y)}{N_2O} + 2H_2O$$

Q.2 Answer:

almost pure p-orbitals are used for $M - H$ bonding.

Solution:

With the decrease in the electronegativity of central atom the bond angle decreases

Q.3 Answer:

passing air through warm acid.

Solution:

Nitric acid decomposes to give NO_2 which is brown

$4HNO_3 \rightarrow 4NO_2 + 2H_2O + O_2$

When air is passed through acid the reaction proceeds towards left hand side and brown colour diminishes.

Q.4 Answer:

NO and O_2

Solution:

$2NO + O_2 \rightarrow 2NO_2$ **brown**

Q.5 Answer:

$BiCl_5$

Solution:

Due to inert pair effect the lower O.S. is more common at the bottom of group

Q.6 Answer:

$[Fe(H_2O)_5 NO]^{2+}$

Solution:

$[Fe(H_2O)_5 NO]^{2+}$ **ion is formed**

Q.7 Answer:

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N

Solution:

Nitrogen can form NCl_3, $N_2O_5Ca_3$ N_2 and not NCl_5 since it has no d atomic orbitals in valence shell

Q.8 Answer:

Catenation tendency is higher in nitrogen when compared with other elements of the same group.

Solution:

Catenation tendency is higher in phosphorus when compared with other elements of same group.

Q.9 Answer:

BCl_3 has no lone pair of electrons but NCl_3 has a lone pair of electrons.

Solution:

In BCl_3 there is sp^2 hybridisation (planar) whereas in NCl_3 hybridisation is sp^3 hence shape is pyramidal

Q.10 Answer:

Mg_3 N_2

Solution:

$$3Mg + N_2 \xrightarrow{\Delta} Mg_3N_2 \xrightarrow{6H_2O}$$

$$3Mg(OH)_2 + 2NH_3\ ;$$

$$CuSO_4 + 4NH_3 \rightarrow [Cu(NH_3)_4]SO_4$$

deep blue colour

Q.11 Answer:

$(SiH_3)_3$ N

Solution:

In case of $N(SiH_3)_3$, N atom is sp^2 hybridised, the lone pair is present in $2p$ orbital and it is transferred to empty d orbital of Si forming $d\pi - p\pi$ bond. Hence nitrogen with sp^2 hybridization has trigonal planar shape.

Q.12 Answer:

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N_2

Solution:

$$Na + NH_3(g) \xrightarrow{\Delta} \underset{[X]}{NaNH_2}$$

$$\xrightarrow{N_2O} \underset{[Y]}{NaN_3} \xrightarrow{\Delta} \underset{[Z]}{N_2} \uparrow$$

Q.13 Answer:

$NH_4Cl + CaO$

Solution:

(a) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Cr_2O_7 + N_2 + H_2O$

(b) $NH_4Cl + NaNO_2 \xrightarrow{\Delta} NH_4NO_2 + NaCl$

(c) $2NH_4Cl + CaO \xrightarrow{\Delta} CaCl_2 + 2NH_3 \uparrow + H_2O$

(d) $NH_4NO_2 \xrightarrow{\Delta} N_2 \uparrow + 2H_2O$

$Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2 \downarrow$

Q.14 Answer:

NO_2 is diamagnetic.

Solution:

NO_2 is paramagnetic and readily dimerises to N_2O_4 which is diamagnetic.

Q.15 Answer:

N_2H_4

Solution:

$2NH_3 + OCl^- \rightarrow NH_2NH_2 + H_2O + Cl^-$

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Q.16 Answer:

Heating NH_4NO_2

Solution:

$$(NH_4)_2\,C_2O_7 \xrightarrow{\Delta} N_2\uparrow + C_2O_3 + 4H_2O$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2\uparrow + 2H_2O$$

Q.17 Answer:

N

Solution:

Only Nitrogen cannot form pentahalide.

Q.18 Answer:

dehydrated

Solution:

$$C_{12}H_{22}O_{11} \xrightarrow[\Delta]{H_2SO_4} \underset{\text{black}}{12C} + 11H_2O$$

Q.19 Answer:

N_2O

Solution:

N_2O_3, N_2O_5 **and** NO_2 **are acidic oxide while** N_2O **is a neutral oxide.**

Q.20 Answer:

NO_2 is formed in (II) reaction

Solution:

Due to the small size of Li^+**, it is very polarizing and favours the formation of** LiO_2**.**

$$4LiO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$$

While other alkali metal nitrates decomposes to give corresponding nitrite. $NaNO_3 \xrightarrow{\Delta} NaNO_2 + \frac{1}{2}O_2$

$$\underset{\substack{\text{white crystalline}\\ \text{solid}}}{\underbrace{4NaNO_3}} \xrightarrow{800^\circ C} 2Na_2O + 2\,N_2 + 5O_2$$

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